

NewAnalysis

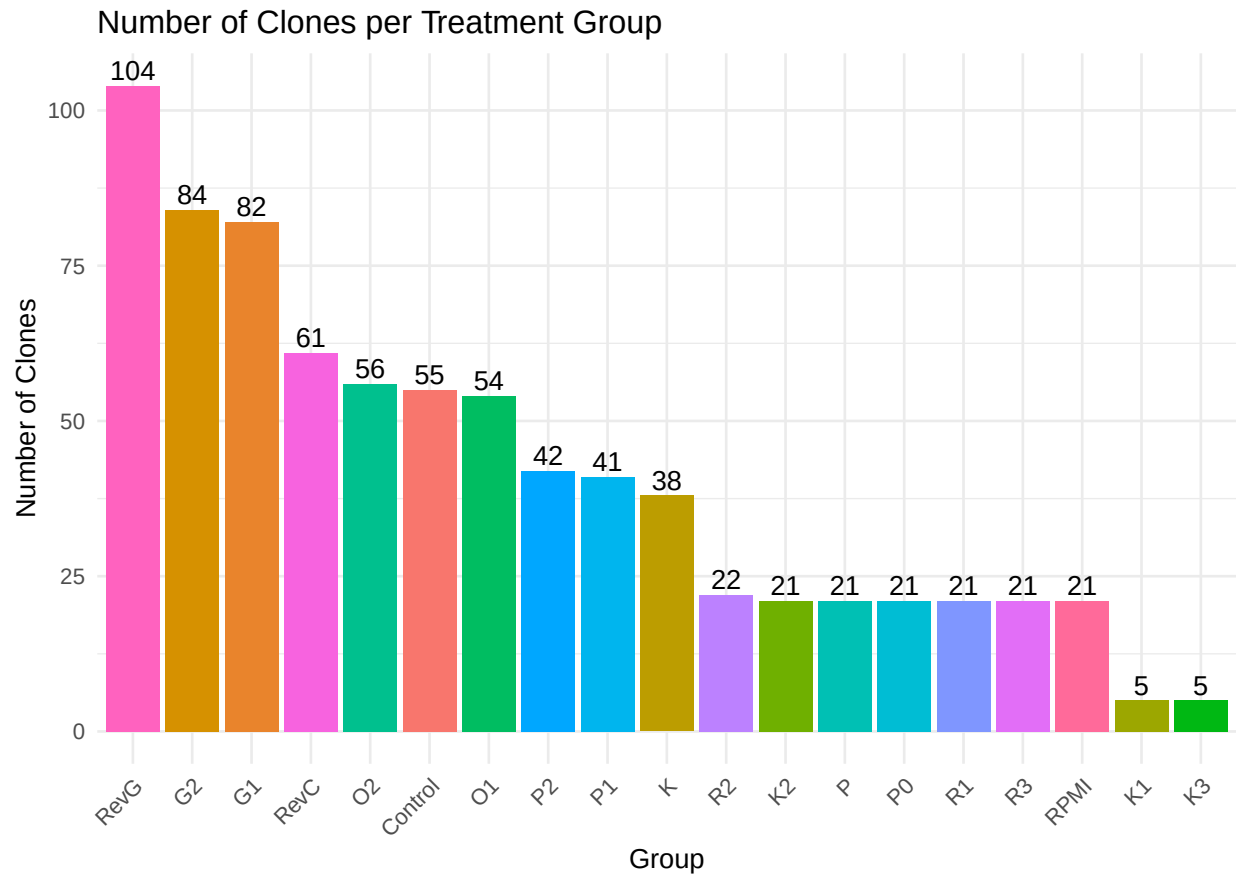
Olofin Daniel

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Objective: To determine whether different treatment conditions induce significant changes in chromosomal copy number profiles across 22 chromosomes among clonal cell populations.

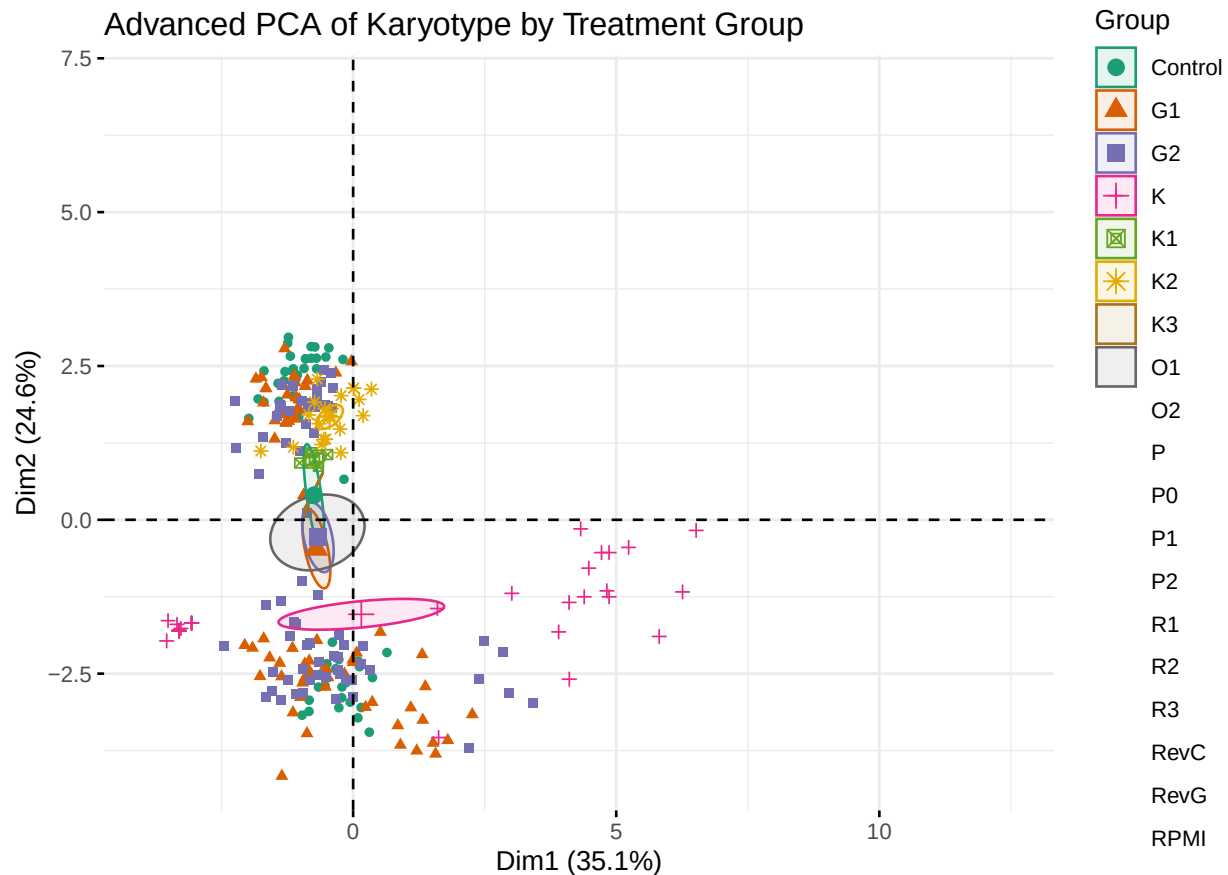
```
# Number of clones per group
group_counts <- karyo_df %>%
  group_by(group) %>%
  summarise(n_clones = n()) %>%
  arrange(desc(n_clones))

# Barplot of group sizes
ggplot(group_counts, aes(x = reorder(group, -n_clones), y = n_clones, fill = group)) +
  geom_bar(stat = "identity", show.legend = F) +
  theme_minimal() +
  labs(title = "Number of Clones per Treatment Group",
       x = "Group",
       y = "Number of Clones") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  geom_text(aes(label = n_clones), vjust = -0.3)
```



```
# PCA using FactoMineR
pca_fm <- PCA(karyo_df[1:22], scale.unit = TRUE, graph = FALSE)

# PCA plot with ellipses for group separation
fviz_pca_ind(pca_fm,
  geom.ind = "point",
  col.ind = karyo_df$group,
  palette = "Dark2",
  addEllipses = TRUE,
  ellipse.type = "confidence",
  legend.title = "Group",
  repel = TRUE) +
  ggtitle("Advanced PCA of Karyotype by Treatment Group")
```

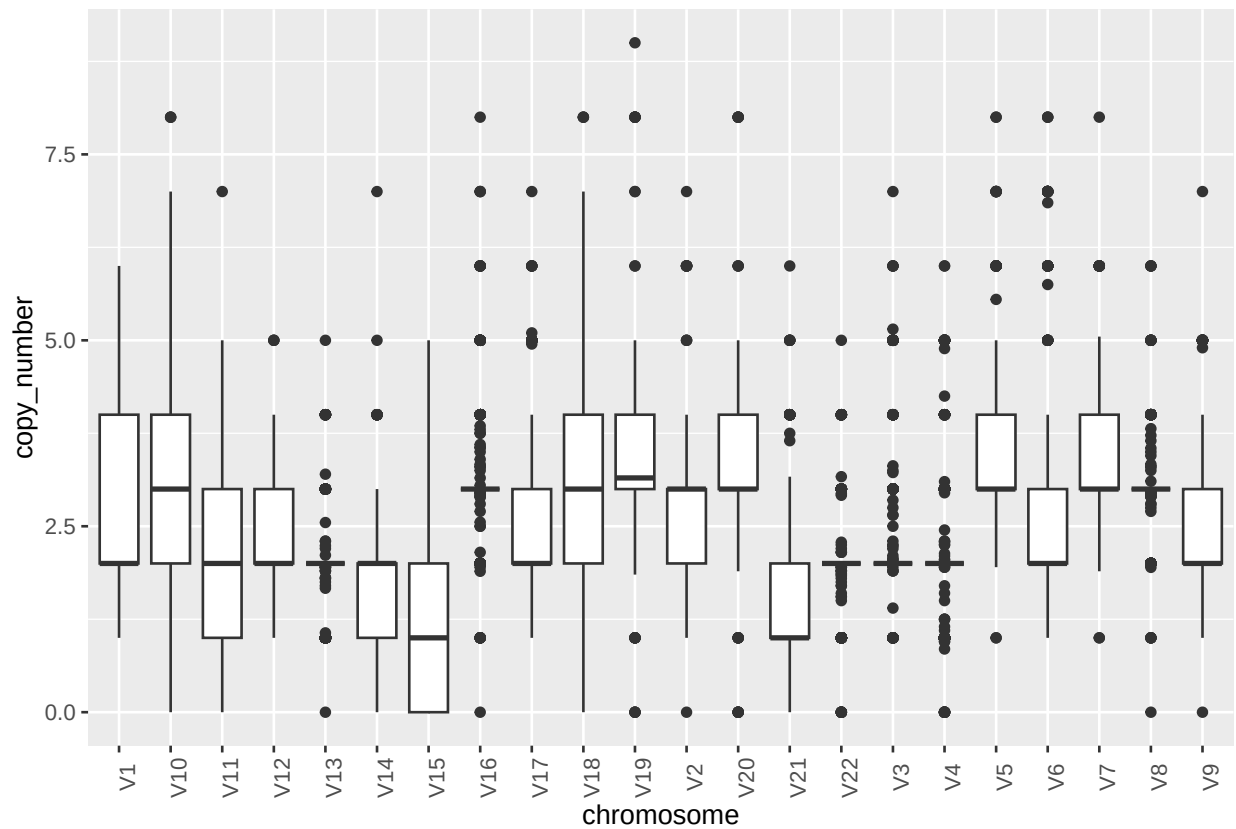


Interpretation: The PCA plot (Dim1: 35.1%, Dim2: 24.6%) reveals distinct clustering patterns in karyotype profiles across treatment groups. Most groups—including Control, G1, G2, K1–K3, and O1–O2—cluster tightly near the origin, indicating similar karyotypic features. In contrast, the K group separates clearly along Dim1 and Dim2, suggesting significant karyotypic alterations in response to treatment. Overall, Dim1 and Dim2 together explain 59.7% of the total variance in the dataset.

```
# Reshape to long format
karyo_long <- karyo_df %>%
  pivot_longer(cols = V1:V22, names_to = "chromosome", values_to = "copy_number")

# Boxplot by chromosome
ggplot(karyo_long, aes(x = chromosome, y = copy_number)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 90)) +
  ggtitle("Distribution of copy number per chromosome")
```

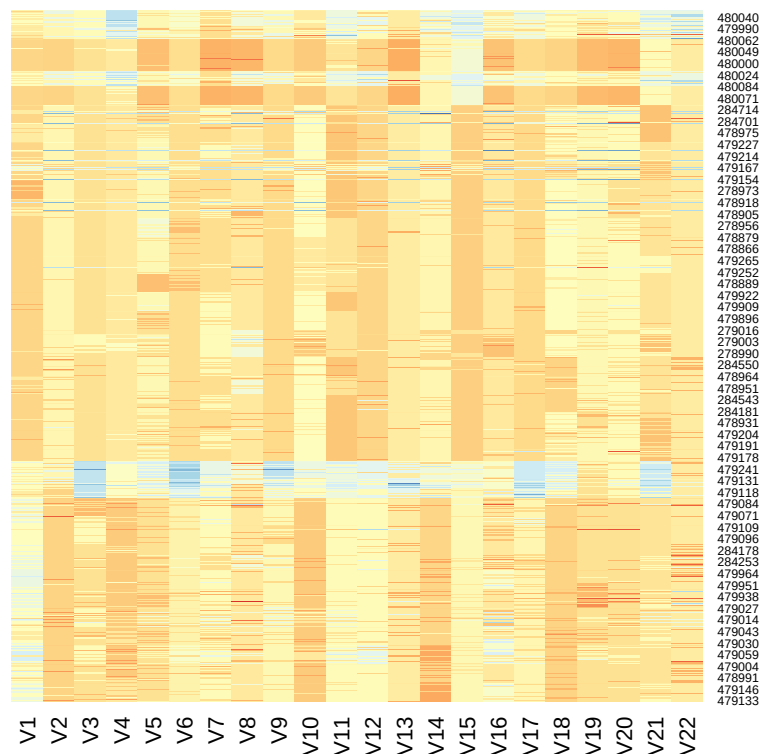
Distribution of copy number per chromosome



Boxplot showing the copy number by chromosome

```
# Prepare matrix for heatmap
karyo_mat <- as.matrix(karyo_df[1:22])
rownames(karyo_mat) <- karyo_df$cloneID
# Z-score normalize across chromosomes (optional)
karyo_scaled <- scale(karyo_mat)
library(RColorBrewer)
# Heatmap
heatmap(karyo_scaled,
  Colv = NA,
  Rowv = NA,
  scale = "none",
  col = colorRampPalette(brewer.pal(9, "RdYlBu"))(100),
  main = "Heatmap of Chromosome Copy Number per Clone")
```

Heatmap of Chromosome Copy Number per Clone

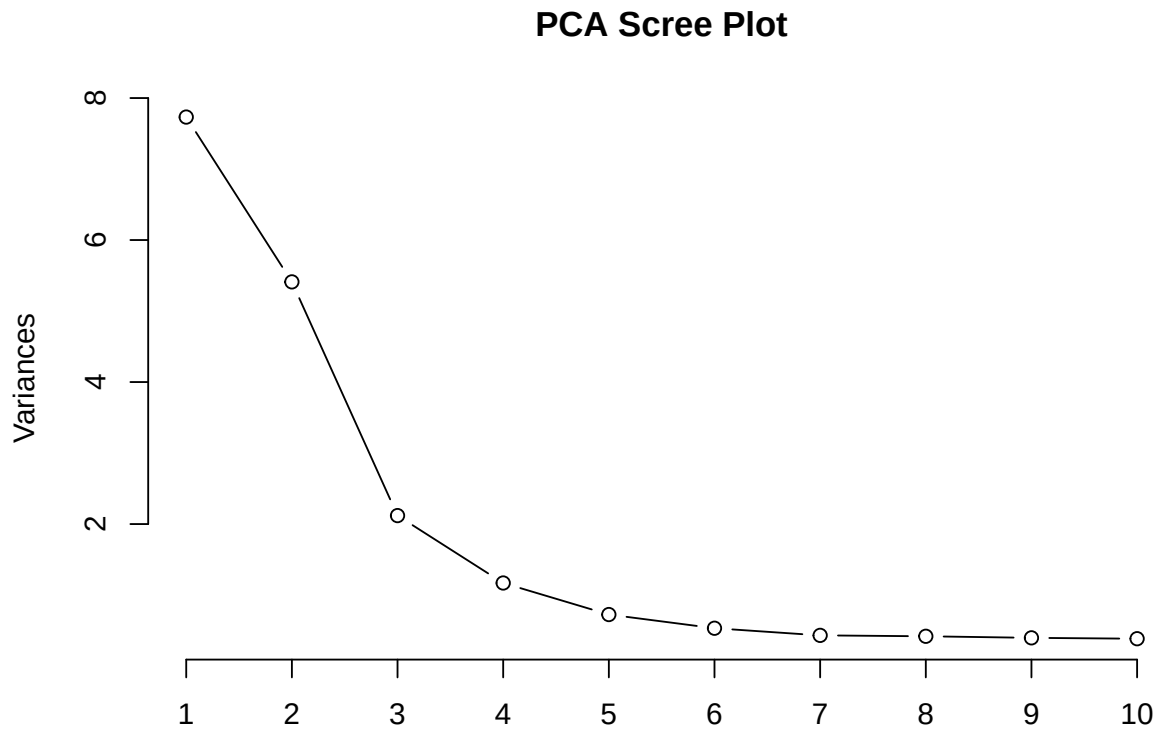


Observation: The heatmap displays z-score normalized chromosome copy numbers across clones (rows) and chromosomes 1-22 (columns). Variations in color intensity indicate differences in chromosomal copy number, with red suggesting gains and blue indicating losses. Certain chromosomes, such as V4, V8, and V9, show greater variability across clones, suggesting regions of chromosomal instability.

```
# Prepare matrix for PCA
karyo_mat <- as.matrix(karyo_df[1:22] )

# PCA
pca_res <- prcomp(karyo_mat, scale. = TRUE)

# Scree plot
plot(pca_res, type = "lines", main = "PCA Scree Plot")
```



Interpretation: The scree plot shows that the first three principal components explain the majority of the variance in the dataset, with a clear drop after the third component. This suggests that PC1 to PC3 capture the most meaningful karyotypic variation, while subsequent components contribute minimally and likely represent noise.

Hypothesis:

Null Hypothesis (H_0): There is no significant difference in karyotype (chromosomal copy number profiles) among clonal populations subjected to different treatment conditions.

Alternative Hypothesis (H_a): There is significant difference in karyotype (chromosomal copy number profiles) among clonal populations subjected to different treatment conditions.

```
# PERMANOVA
```

```
adonis2(karyo_mat ~ group, data = karyo_df, method = "euclidean", permutations = 999)
```

```
## Permutation test for adonis under reduced model
```

```
## Permutation: free
```

```
## Number of permutations: 999
```

```
##
```

```
## adonis2(formula = karyo_mat ~ group, data = karyo_df, permutations = 999, method = "euclidean")
```

```
##           Df SumOfSqs    R2      F Pr(>F)
```

```
## Model      18   5170.6 0.351 22.715  0.001 ***
```

```
## Residual  756   9560.2 0.649
```

```
## Total     774  14730.8 1.000
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation: A permutational multivariate analysis of variance (PERMANOVA) using Euclidean distance revealed a statistically significant difference in overall karyotype profiles across the 19 treatment groups ($F = 22.72$, $R^2 = 0.351$, $p = 0.001$). This suggests that treatment condition accounts for ~35% of the variance in chromosome-level copy number. However, a test for homogeneity of dispersion (PERMDISP) indicated significant variance heterogeneity among groups ($p < 0.05$), implying that the observed effect may partly reflect differences in within-group variability. These results highlight both treatment-induced karyotype shifts and potential genomic instability under specific conditions.

```
# Check beta-dispersion (group variances)
dist_mat <- dist(karyo_mat)
beta_disp <- betadisper(dist_mat, karyo_df$group)

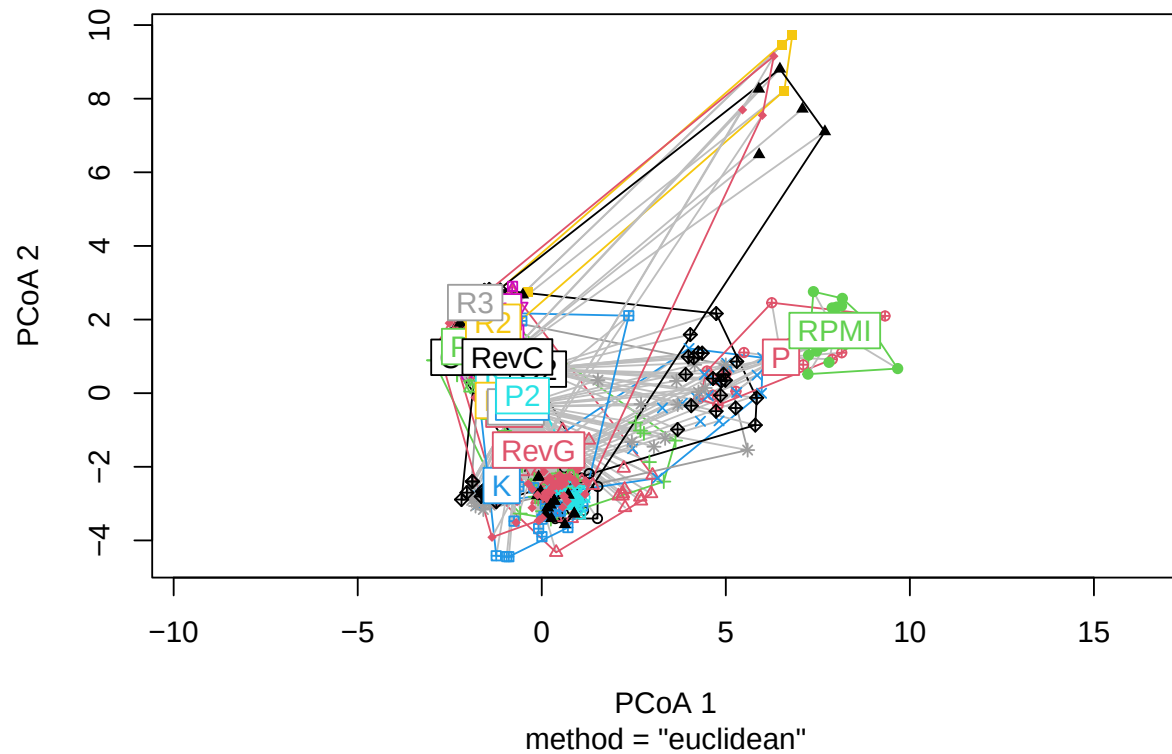
# Test significance of dispersion differences
permutest(beta_disp)
```

```
##
## Permutation test for homogeneity of multivariate dispersions
## Permutation: free
## Number of permutations: 999
##
## Response: Distances
##           Df  Sum Sq Mean Sq      F N.Perm Pr(>F)
## Groups      18   574.12  31.896 11.408   999  0.001 ***
## Residuals 756  2113.64   2.796
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation: The test for homogeneity of multivariate dispersions (PERMDISP) was statistically significant ($F = 11.41$, $p = 0.001$), indicating that group variances differ in multivariate space. This suggests that the PERMANOVA result ($F = 22.72$, $p = 0.001$) should be interpreted with caution, as it may reflect differences in within-group variability as well as differences in centroid (mean) karyotype profiles. Follow-up pairwise dispersion tests are underway to determine which groups contribute most to this variance.

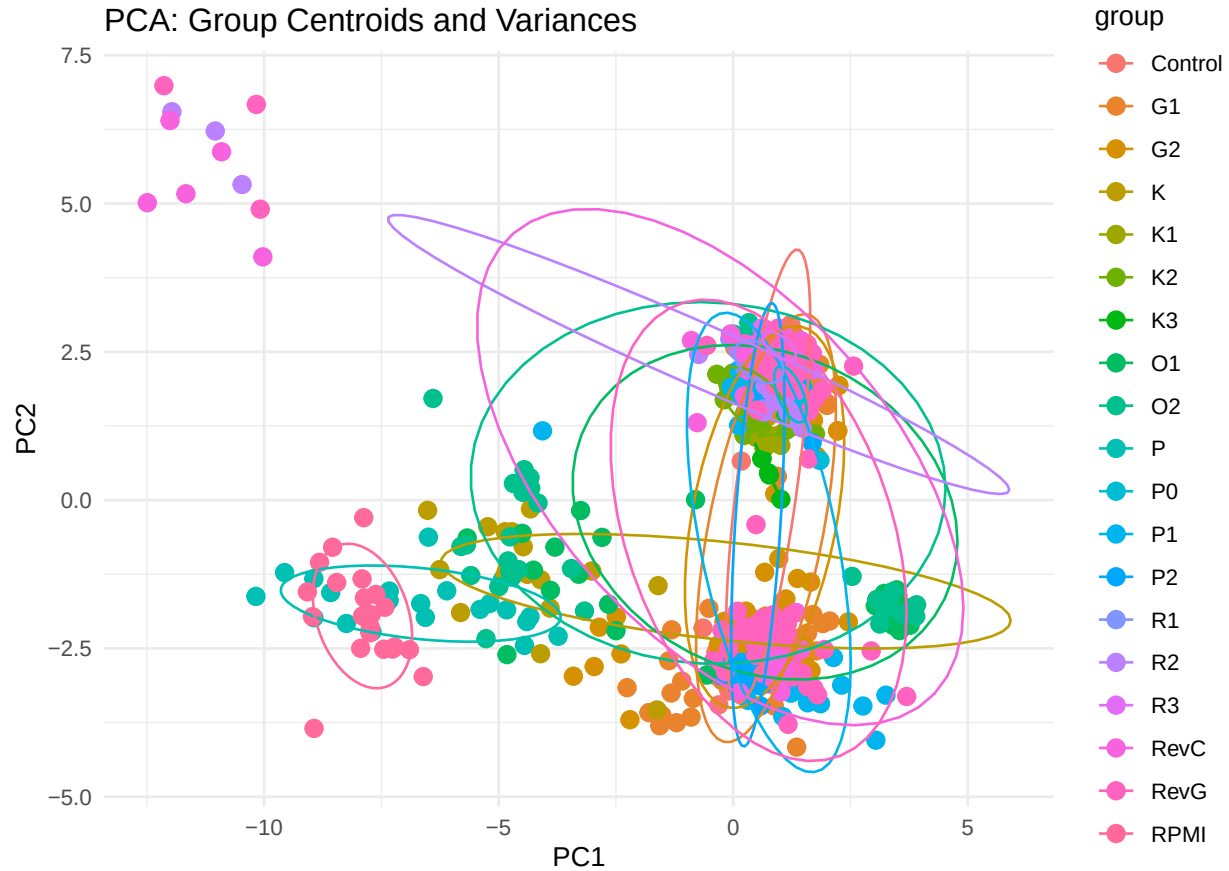
```
# Plot
plot(beta_disp, main = "Group Dispersion (Variance around Group Centroid)")
```

Group Dispersion (Variance around Group Centroid)



```
# PCA coordinates from base prcomp()
pca <- prcomp(karyo_df[, 1:22], scale. = TRUE)
pca_df <- data.frame(pca$x[, 1:2], group = karyo_df$group)

# PCA plot
ggplot(pca_df, aes(x = PC1, y = PC2, color = group)) +
  geom_point(size = 3) +
  stat_ellipse(type = "norm", level = 0.68) +
  theme_minimal() +
  ggtitle("PCA: Group Centroids and Variances")
```



```
TukeyHSD(beta_disp)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = distances ~ group, data = df)
##
## $group
##
```

	diff	lwr	upr	p adj
G1-Control	0.353852902	-0.67476025	1.38246605	0.9995093
G2-Control	0.096105690	-0.92757962	1.11979100	1.0000000
K-Control	0.568371586	-0.67656833	1.81331150	0.9846531
K1-Control	-2.196063617	-4.95276255	0.56063531	0.3288335
K2-Control	-1.271177751	-2.78507313	0.24271763	0.2376173
K3-Control	-2.523583170	-5.28028210	0.23311576	0.1214182
O1-Control	0.755453981	-0.37516191	1.88606987	0.6669425
O2-Control	1.242821281	0.12243911	2.36320346	0.0131147
P-Control	0.195806194	-1.31808918	1.70970157	1.0000000
P0-Control	-2.251302111	-3.76519749	-0.73740673	0.0000329
P1-Control	0.475266532	-0.74243935	1.69297241	0.9974860
P2-Control	-0.002588333	-1.21196041	1.20678374	1.0000000
R1-Control	-1.832528449	-3.34642383	-0.31863307	0.0032002
R2-Control	-0.051135076	-1.53992265	1.43765250	1.0000000
R3-Control	-2.305105301	-3.81900068	-0.79120992	0.0000170
RevC-Control	0.769512704	-0.32788183	1.86690724	0.5777227

## RevG-Control	0.189234613	-0.79473255	1.17320178	0.9999999
## RPMI-Control	-0.677329745	-2.19122512	0.83656563	0.9876577
## G2-G1	-0.257747212	-1.17394140	0.65844698	0.9999719
## K-G1	0.214518684	-0.94365037	1.37268774	1.0000000
## K1-G1	-2.549916519	-5.26853182	0.16869878	0.0971895
## K2-G1	-1.625030653	-3.06841536	-0.18164595	0.0105233
## K3-G1	-2.877436072	-5.59605137	-0.15882077	0.0250762
## O1-G1	0.401601078	-0.63269697	1.43589913	0.9976420
## O2-G1	0.888968379	-0.13413298	1.91206974	0.1861046
## P-G1	-0.158046708	-1.60143141	1.28533799	1.0000000
## P0-G1	-2.605155013	-4.04853971	-1.16177031	0.0000000
## P1-G1	0.121413630	-1.00742993	1.25025719	1.0000000
## P2-G1	-0.356441235	-1.47628988	0.76340741	0.9998319
## R1-G1	-2.186381351	-3.62976605	-0.74299665	0.0000197
## R2-G1	-0.404987978	-1.82201608	1.01204012	0.9999645
## R3-G1	-2.658958203	-4.10234291	-1.21557350	0.0000000
## RevC-G1	0.415659802	-0.58221541	1.41353501	0.9944979
## RevG-G1	-0.164618289	-1.03620969	0.70697311	0.9999999
## RPMI-G1	-1.031182647	-2.47456735	0.41220205	0.5414718
## K-G2	0.472265896	-0.68152879	1.62606058	0.9955240
## K1-G2	-2.292169307	-5.00892395	0.42458533	0.2299783
## K2-G2	-1.367283442	-2.80716052	0.07259364	0.0862843
## K3-G2	-2.619688860	-5.33644350	0.09706578	0.0739702
## O1-G2	0.659348290	-0.37004914	1.68874572	0.7372150
## O2-G2	1.146715591	0.12856874	2.16486244	0.0104628
## P-G2	0.099700504	-1.34017658	1.53957758	1.0000000
## P0-G2	-2.347407801	-3.78728488	-0.90753072	0.0000021
## P1-G2	0.379160842	-0.74519426	1.50351594	0.9996255
## P2-G2	-0.098694023	-1.21401801	1.01662997	1.0000000
## R1-G2	-1.928634139	-3.36851122	-0.48875706	0.0004285
## R2-G2	-0.147240766	-1.56069584	1.26621431	1.0000000
## R3-G2	-2.401210991	-3.84108807	-0.96133391	0.0000010
## RevC-G2	0.673407014	-0.31938780	1.66620183	0.6400445
## RevG-G2	0.093128923	-0.77264135	0.95889920	1.0000000
## RPMI-G2	-0.773435435	-2.21331251	0.66644164	0.9266134
## K1-K	-2.764435203	-5.57204977	0.04317936	0.0593543
## K2-K	-1.839549338	-3.44428912	-0.23480955	0.0079687
## K3-K	-3.091954756	-5.89956932	-0.28434019	0.0145647
## O1-K	0.187082394	-1.06255869	1.43672348	1.0000000
## O2-K	0.674449695	-0.56594007	1.91483946	0.9186560
## P-K	-0.372565392	-1.97730518	1.23217439	0.9999985
## P0-K	-2.819673697	-4.42441348	-1.21493391	0.0000002
## P1-K	-0.093105054	-1.42205884	1.23584873	1.0000000
## P2-K	-0.570959919	-1.89228175	0.75036191	0.9916104
## R1-K	-2.400900035	-4.00563982	-0.79616025	0.0000279
## R2-K	-0.619506662	-2.20058193	0.96156861	0.9973617
## R3-K	-2.873476887	-4.47821667	-1.26873710	0.0000001
## RevC-K	0.201141118	-1.01852495	1.42080718	1.0000000
## RevG-K	-0.379136973	-1.49784245	0.73956850	0.9995992
## RPMI-K	-1.245701331	-2.85044111	0.35903845	0.3777483
## K2-K1	0.924885866	-2.01189866	3.86167040	0.9998553
## K3-K1	-0.327519553	-4.06010755	3.40506844	1.0000000
## O1-K1	2.951517598	0.19269241	5.71034279	0.0217593
## O2-K1	3.438884898	0.68423785	6.19353195	0.0018144

## P-K1	2.391869811	-0.54491472	5.32865434	0.2889316
## P0-K1	-0.055238494	-2.99202302	2.88154604	1.0000000
## P1-K1	2.671330150	-0.12431499	5.46697529	0.0811079
## P2-K1	2.193475284	-0.59854996	4.98550053	0.3548594
## R1-K1	0.363535168	-2.57324936	3.30031970	1.0000000
## R2-K1	2.144928542	-0.77899222	5.06884930	0.4891517
## R3-K1	-0.109041684	-3.04582621	2.82774285	1.0000000
## RevC-K1	2.965576321	0.22019859	5.71095406	0.0190963
## RevG-K1	2.385298230	-0.31674091	5.08733737	0.1646325
## RPMI-K1	1.518733872	-1.41805066	4.45551840	0.9474525
## K3-K2	-1.252405418	-4.18918995	1.68437911	0.9927778
## 01-K2	2.026631732	0.50886803	3.54439544	0.0004589
## 02-K2	2.513999033	1.00384321	4.02415485	0.0000011
## P-K2	1.466983946	-0.35433250	3.28830039	0.3087181
## P0-K2	-0.980124359	-2.80144081	0.84119209	0.9254567
## P1-K2	1.746444284	0.16273914	3.33014943	0.0142855
## P2-K2	1.268589419	-0.30871689	2.84589573	0.3113602
## R1-K2	-0.561350698	-2.38266715	1.25996575	0.9998935
## R2-K2	1.220042676	-0.58045804	3.02054340	0.6418711
## R3-K2	-1.033927550	-2.85524400	0.78738890	0.8849197
## RevC-K2	2.040690455	0.54750957	3.53387134	0.0002724
## RevG-K2	1.460412364	0.04849673	2.87232800	0.0335665
## RPMI-K2	0.593848006	-1.22746844	2.41516445	0.9997645
## 01-K3	3.279037150	0.52021196	6.03786234	0.0044003
## 02-K3	3.766404451	1.01175740	6.52105150	0.0002696
## P-K3	2.719389364	-0.21739517	5.65617389	0.1095417
## P0-K3	0.272281059	-2.66450347	3.20906559	1.0000000
## P1-K3	2.998849702	0.20320456	5.79449484	0.0209981
## P2-K3	2.520994837	-0.27103041	5.31302008	0.1367954
## R1-K3	0.691054721	-2.24572981	3.62783925	0.9999982
## R2-K3	2.472448094	-0.45147267	5.39636885	0.2264864
## R3-K3	0.218477868	-2.71830666	3.15526240	1.0000000
## RevC-K3	3.293095874	0.54771814	6.03847361	0.0037565
## RevG-K3	2.712817783	0.01077864	5.41485693	0.0477907
## RPMI-K3	1.846253424	-1.09053110	4.78303795	0.7653836
## 02-01	0.487367301	-0.63823639	1.61297099	0.9914177
## P-01	-0.559647786	-2.07741149	0.95811592	0.9987738
## P0-01	-3.006756091	-4.52451980	-1.48899238	0.0000000
## P1-01	-0.280187448	-1.50269923	0.94232433	0.9999988
## P2-01	-0.758042313	-1.97225328	0.45616865	0.7753850
## R1-01	-2.587982430	-4.10574614	-1.07021872	0.0000005
## R2-01	-0.806589056	-2.29931003	0.68613192	0.9228239
## R3-01	-3.060559282	-4.57832299	-1.54279558	0.0000000
## RevC-01	0.014058723	-1.08866618	1.11678363	1.0000000
## RevG-01	-0.566219368	-1.55612787	0.42368913	0.8780327
## RPMI-01	-1.432783726	-2.95054743	0.08497998	0.0913762
## P-02	-1.047015087	-2.55717091	0.46314073	0.5993516
## P0-02	-3.494123392	-5.00427921	-1.98396757	0.0000000
## P1-02	-0.767554749	-1.98060833	0.44549883	0.7558387
## P2-02	-1.245409614	-2.45009721	-0.04072202	0.0337851
## R1-02	-3.075349730	-4.58550555	-1.56519391	0.0000000
## R2-02	-1.293956357	-2.77894115	0.19102844	0.1821532
## R3-02	-3.547926583	-5.05808240	-2.03777076	0.0000000
## RevC-02	-0.473308577	-1.56553847	0.61892132	0.9913373

## RevG-O2	-1.053586668	-2.03179052	-0.07538282	0.0198718
## RPMI-O2	-1.920151026	-3.43030685	-0.40999521	0.0012715
## P0-P	-2.447108305	-4.26842475	-0.62579186	0.0004002
## P1-P	0.279460338	-1.30424481	1.86316549	1.0000000
## P2-P	-0.198394527	-1.77570084	1.37891179	1.0000000
## R1-P	-2.028334643	-3.84965109	-0.20701819	0.0123782
## R2-P	-0.246941270	-2.04744199	1.55355945	1.0000000
## R3-P	-2.500911495	-4.32222794	-0.67959505	0.0002443
## RevC-P	0.573706510	-0.91947437	2.06688739	0.9979319
## RevG-P	-0.006571581	-1.41848722	1.40534406	1.0000000
## RPMI-P	-0.873135939	-2.69445239	0.94818051	0.9745449
## P1-P0	2.726568643	1.14286350	4.31027379	0.0000003
## P2-P0	2.248713778	0.67140747	3.82602009	0.0000989
## R1-P0	0.418773662	-1.40254279	2.24009011	0.9999987
## R2-P0	2.200167035	0.39966632	4.00066775	0.0027000
## R3-P0	-0.053803191	-1.87511964	1.76751326	1.0000000
## RevC-P0	3.020814815	1.52763393	4.51399570	0.0000000
## RevG-P0	2.440536724	1.02862108	3.85245236	0.0000003
## RPMI-P0	1.573972366	-0.24734408	3.39528881	0.1938107
## P2-P1	-0.477854865	-1.77354912	0.81783939	0.9987708
## R1-P1	-2.307794981	-3.89150013	-0.72408983	0.0000565
## R2-P1	-0.526401608	-2.08612313	1.03331992	0.9996213
## R3-P1	-2.780371834	-4.36407698	-1.19666669	0.0000002
## RevC-P1	0.294246172	-0.89760849	1.48610083	0.9999962
## RevG-P1	-0.286031920	-1.37434904	0.80228520	0.9999899
## RPMI-P1	-1.152596278	-2.73630142	0.43110887	0.5048962
## R1-P2	-1.829940116	-3.40724643	-0.25263380	0.0065787
## R2-P2	-0.048546743	-1.60177063	1.50467714	1.0000000
## R3-P2	-2.302516969	-3.87982328	-0.72521066	0.0000540
## RevC-P2	0.772101037	-0.41123777	1.95543985	0.7075277
## RevG-P2	0.191822945	-0.88716147	1.27080736	1.0000000
## RPMI-P2	-0.674741413	-2.25204773	0.90256490	0.9925171
## R2-R1	1.781393374	-0.01910735	3.58189409	0.0563077
## R3-R1	-0.472576852	-2.29389330	1.34873960	0.9999917
## RevC-R1	2.602041153	1.10886027	4.09522203	0.0000002
## RevG-R1	2.021763062	0.60984742	3.43367870	0.0000886
## RPMI-R1	1.155198704	-0.66611774	2.97651515	0.7522421
## R3-R2	-2.253970226	-4.05447095	-0.45346951	0.0017210
## RevC-R2	0.820647780	-0.64707099	2.28836655	0.8978444
## RevG-R2	0.240369688	-1.14459060	1.62532998	1.0000000
## RPMI-R2	-0.626194670	-2.42669539	1.17430605	0.9994316
## RevC-R3	3.074618005	1.58143712	4.56779889	0.0000000
## RevG-R3	2.494339914	1.08242428	3.90625555	0.0000001
## RPMI-R3	1.627775556	-0.19354089	3.44909200	0.1490132
## RevG-RevC	-0.580278091	-1.53206657	0.37151039	0.8074743
## RPMI-RevC	-1.446842449	-2.94002333	0.04633843	0.0703557
## RPMI-RevG	-0.866564358	-2.27848000	0.54535128	0.7987747

Interpretation: Pairwise tests for multivariate dispersion revealed that several treatment groups — notably P0, R3, and O2 — exhibited significantly higher or lower within-group variability compared to others (e.g., Control, G1, G2). These findings indicate that the significant PERMANOVA result ($p < 0.001$) may be partially attributable to heterogeneity in dispersion, not just shifts in mean karyotype profiles. Consequently, group differences involving these high-dispersion treatments should be interpreted with caution, as they may reflect increased genomic instability rather than systematic karyotype restructuring.

```
# Prepare karyotype matrix
karyo_mat <- as.matrix(karyo_df[, 1:22])
```

```
# Step 1: Compute distance matrix
dist_mat <- dist(karyo_mat)
```

```
# Step 2: Test for homogeneity of group dispersions
dispersion <- betadisper(dist_mat, karyo_df$group)
disp_result <- permutest(dispersion)
print(disp_result)
```

```
##
## Permutation test for homogeneity of multivariate dispersions
## Permutation: free
## Number of permutations: 999
##
## Response: Distances
##          Df  Sum Sq Mean Sq      F N.Perm Pr(>F)
## Groups    18  574.12  31.896 11.408   999 0.001 ***
## Residuals 756 2113.64   2.796
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Step 3: Pairwise Tukey HSD test
tukey_disp <- TukeyHSD(dispersion)
```

```
# Step 4: Identify groups with equal dispersion
# Extract pairwise adjusted p-values
tukey_df <- as.data.frame(tukey_disp$group)
tukey_df$Comparison <- rownames(tukey_df)
```

```
# Filter for non-significant comparisons (p > 0.1)
equal_disp_pairs <- tukey_df %>% filter(`p adj` > 0.1)
```

```
# Extract all groups from non-significant comparisons
safe_groups <- unique(unlist(strsplit(equal_disp_pairs$Comparison, "-")))
```

```
# Step 5: Subset original data to "safe" groups only
subset_df <- karyo_df %>% filter(group %in% safe_groups)
subset_mat <- as.matrix(subset_df[, 1:22])
```

```
# Step 6: Run PERMANOVA on subset
adonis2(subset_mat ~ group, data = subset_df, method = "euclidean", permutations = 999)
```

```
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = subset_mat ~ group, data = subset_df, permutations = 999, method = "euclidean")
##          Df SumOfSqs    R2      F Pr(>F)
## Model    18  5170.6 0.351 22.715 0.001 ***
## Residual 756  9560.2 0.649
```

```
## Total      774  14730.8 1.000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation: To ensure that our PERMANOVA results were not confounded by differences in group dispersion, we conducted a pairwise Tukey HSD test following a significant PERMDISP result. We identified a subset of groups with homogeneous multivariate dispersion and re-ran PERMANOVA on this subset. The result remained highly significant ($F = 22.72$, $R^2 = 0.351$, $p = 0.001$), confirming that treatment-associated differences in karyotype are robust and reflect true shifts in multivariate profile, rather than just variance heterogeneity.

```
adonis2(karyo_mat ~ group, data = karyo_df, method = "bray")
```

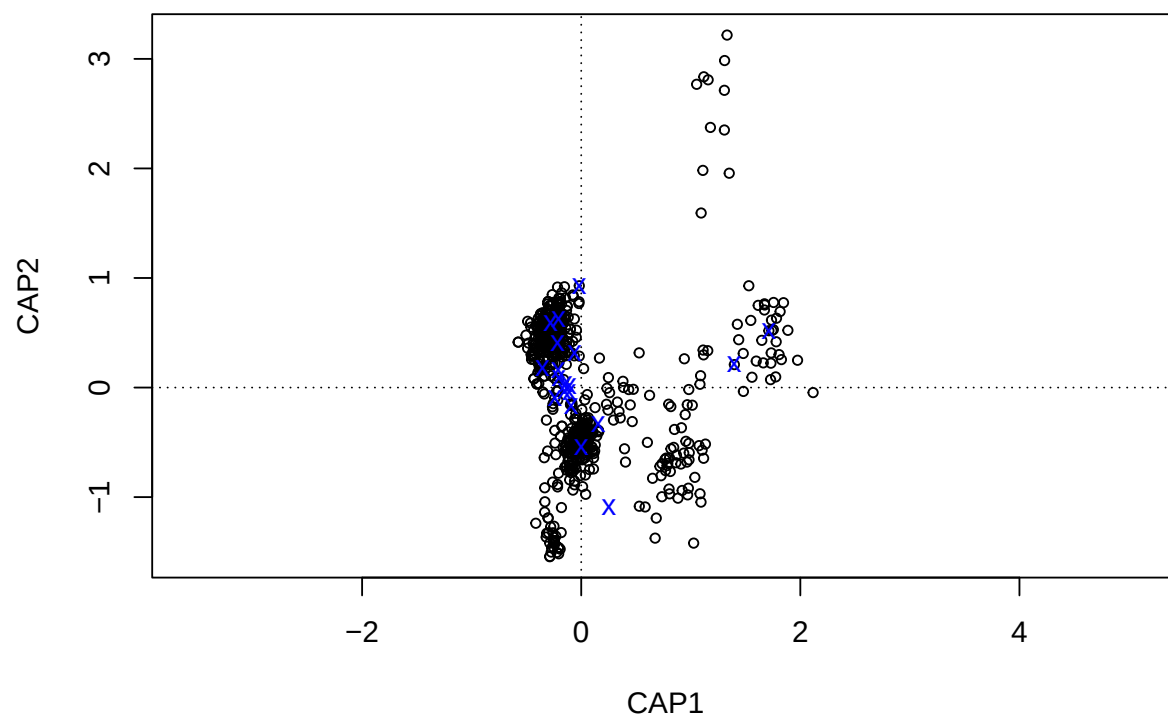
```
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = karyo_mat ~ group, data = karyo_df, method = "bray")
##           Df SumOfSqs      R2      F Pr(>F)
## Model      18   4.2680 0.31251 19.092  0.001 ***
## Residual  756   9.3893 0.68749
## Total     774  13.6574 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
cap <- capscale(dist_mat ~ group, data = karyo_df)
anova(cap, permutations = 999)
```

```
## Permutation test for capscale under reduced model
## Permutation: free
## Number of permutations: 999
##
## Model: capscale(formula = dist_mat ~ group, data = karyo_df)
##           Df Variance      F Pr(>F)
## Model      18   6.6803 22.715  0.001 ***
## Residual  756  12.3517
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Intepretation: A constrained analysis of principal coordinates (CAP) was conducted to visualize and confirm group-level differences in karyotype structure. The result was highly significant ($F = 22.72$, $p = 0.001$), with treatment groups explaining approximately 35.1% of the total variation. This reinforces the findings from PERMANOVA and provides strong multivariate evidence that treatment groups differ in chromosomal copy number profiles.

```
plot(cap)
```



Interpretation: The CAP (Canonical Analysis of Principal coordinates) plot displays the distribution of samples along the first two canonical axes (CAP1 and CAP2). Most samples cluster tightly around the origin, indicating moderate variation in the constrained ordination space. A few points spread out along CAP1, suggesting some treatment or group differences. Blue “X” markers likely represent group centroids or means, which show partial separation along CAP1. This suggests there may be subtle but meaningful differences in karyotype composition across treatment groups.

Statistical Workflow: Karyotype Analysis by Treatment Group

